MARKETING CAMPAIGN EFFECTIVENESS ANALYSIS

Consumer Behaviour, Advertising Trust, and Promotional Preferences

Data Handling and Visualization — IDHV 2026M

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Abstract

This project analyses self-reported consumer behaviour toward online advertising and promotional offers using a cleaned survey dataset of 80 respondents. The questionnaire covers age group, product-discovery platforms, advertising exposure, ad-click and purchase-after-ad behaviour, trust in advertising, discount importance, preferred incentives, and purchase intention. The analysis uses descriptive statistics, category distributions, rating-distribution charts, a Spearman correlation heatmap, box plots, and cross-tabulations. The supplied analysis dataset is respondent-level and does not contain the three scenario-level ratings from the promotional experiment; therefore, the report documents the experimental design but does not invent a result comparing immediate discount, cashback, and no-promotion purchase-intention scores. Findings describe this sample and should not be generalized to all consumers.

Executive Summary

The cleaned file contains 80 respondent records and 16 columns. The most common product-discovery platform is Instagram (26 respondents; 32.5%). The most preferred promotion is Immediate price discount (33; 41.2%). The most common reported reason for ignoring an advertisement is I prefer researching products independently (34; 42.5%). Mean ad-trust rating is 3.00/5, mean discount-importance rating is 3.33/5, and mean likelihood of choosing a promoted product is 3.55/5. These are descriptive results, not evidence of causal impact.

A key evidence boundary is that the supplied cleaned CSV contains 80 respondent-level records, but no scenario-condition field or scenario-level outcome ratings. The three-condition experiment is described in the collection documentation and screenshots, but the comparative experimental outcome analysis requires the experiment response export.

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1. Introduction and Problem Statement

Online advertisements are widely used to introduce products, communicate offers, and influence consumer decisions. However, seeing an advertisement does not guarantee that a person trusts it, clicks it, or purchases the product. Consumers may prefer to research products independently, may find advertisements irrelevant, or may respond differently to immediate discounts and delayed cashback.

This project studies self-reported advertising behaviour and promotional preferences. It also documents a Streamlit-based experiment that presents the same wireless noise-cancelling headphones under three conditions: a ₹400 immediate discount, ₹400 cashback after paying the original ₹2,000 price, and no promotion. The available cleaned analysis file supports the survey analysis; scenario-level experiment results are not included in that file.

2. Research Questions and Objectives

- Which platforms do respondents most often use to discover products?
- How frequently do respondents encounter online advertisements, and how do they report clicking or purchasing after seeing ads?
- What are respondents' levels of trust in advertising information and sensitivity to discounts?
- Which promotional incentives are most preferred, and what are the main reasons for ignoring ads?
- Are ordinal ratings associated with one another, and how do responses vary across age groups and advertising behaviours?
- How should a future analysis compare the three promotional scenarios while accounting for each participant rating all three?

The main objectives are to summarize respondent characteristics, inspect data quality, visualize important categorical and ordinal variables, examine associations without implying causation, and establish a sound analysis plan for the repeated-measures promotional experiment.

3. Data Collection and Questionnaire

The data collection application was built with Python and Streamlit. The documented workflow uses the app to collect survey responses and stores them in Google Sheets through an API/service-account integration, after which the dataset is exported for Python-based preprocessing and analysis. The screenshots supplied with the project are evidence of the questionnaire design; they are not reproduced in this report.

Age group was the only required field in Part 1. All other survey questions were optional. Question 6a asked which platform showed an advertisement before purchase and was conditional on the purchase-after-ad response according to the survey flow.

3.1 Part 1 — Consumer behaviour questions

Q	Topic	Response format	Requirement
1	Age group	Ordered categorical dropdown	Required
2	Main platform used to discover or learn about products	Categorical dropdown	Optional
3	Frequency of encountering online advertisements	Ordered categorical choice	Optional
4	Most frequently purchased online product category	Categorical dropdown	Optional
5	Clicked an online ad in the past three months	Yes / No / Not sure	Optional
6	Purchased a product after seeing an online ad in the past three months	Yes / No / Not sure	Optional
6a	Platform where the ad was seen before purchasing	Conditional categorical dropdown	Optional / conditional

7	Trust in product information shown in online ads	1–5 rating	Optional
8	Main influence on purchase decision after seeing an ad	Categorical dropdown	Optional
9	Main reason for ignoring an online ad	Categorical dropdown	Optional
10	Importance of a discount when deciding to purchase	1–5 rating	Optional
11	Promotion most likely to encourage consideration	Categorical dropdown	Optional
12	Likelihood of choosing a promoted product if two products are equally suitable	1–5 rating	Optional

3.2 Part 2 — Promotional experiment design

Participants were shown three versions of the same wireless noise-cancelling headphones, one at a time, and were asked to evaluate each scenario independently. The product's base price was ₹2,000. The documented application randomized scenario order to help reduce order effects.

Condition	Offer	Paid at checkout	Effective price after offer
A — Immediate discount	₹400 immediate discount	₹1,600	₹1,600
B — Cashback	₹400 cashback after purchase	₹2,000	₹1,600 after cashback is received
C — No promotion	No discount or cashback	₹2,000	₹2,000

For each condition, the three outcomes were purchase likelihood, attractiveness of the offer/purchase option, and confidence that the option provided good value for money, each rated on a 1–5 scale.

The experiment is a within-subject design because each participant evaluates all three conditions. Valid inferential comparisons should therefore use paired or repeated-measures methods rather than treating all scenario ratings as independent observations.

4. Dataset and Data Dictionary

The selected cleaned dataset, cleaned_data_step2.csv, contains 80 rows and 16 columns. Each row represents one respondent. The file contains respondent-level survey variables and the derived composite score, but it does not contain scenario-level experimental ratings.

Column	Meaning	Type
respondent_id	Respondent identifier	Identifier
timestamp_utc	Survey submission timestamp	Datetime/text
age_group	Respondent age band	Ordinal category
discovery_platform	Main product-discovery platform	Nominal category
ad_frequency	Frequency of encountering online ads	Ordinal category
online_product_category	Most frequently purchased product category	Nominal category
clicked_ad_3m	Clicked an online ad in past three months	Categorical
purchased_after_ad_3m	Purchase after seeing an online ad in past three months	Categorical
purchase_ad_platform	Platform associated with ad before purchase	Conditional category
ad_trust	Trust in ad information (1–5)	Ordinal rating
purchase_influence	Main purchase influence	Nominal category
ad_ignore_reason	Main reason for ignoring ads	Nominal category
discount_importance	Importance of discounts (1–5)	Ordinal rating
preferred_promotion	Preferred promotion type	Nominal category
choose_promoted_product_likelihood	Likelihood of choosing promoted product (1–5)	Ordinal rating
ad_receptivity_score	Mean of the three listed rating constructs	Derived numeric score

5. Preprocessing and Data Quality

The Step 2 preprocessing report documents character/Unicode cleanup, categorical-label trimming and standardization, conversion of the three survey ratings to numeric values, checks that rating values remain within the valid 1–5 domain, replacement of structurally inapplicable purchase-platform responses with “Not applicable,” and removal of the constant administrative status column. The report states that the cleaned respondent dataset retained all 80 rows and that complete core survey fields had no null values at that processing stage.

The updated CSV used here has 80 rows. Its current missingness and duplicate-ID checks are summarized below; these checks were recalculated from the actual file rather than copied without verification from the preprocessing report.

Quality check	Result
Rows / respondents	80
Columns in supplied cleaned file	16
Duplicate respondent IDs	0
Total missing cells	0
Rating variables all within 1–5	Yes
Normalization	Not applied; native 1–5 rating scale preserved
Composite score	Unweighted mean of ad_trust, discount_importance, and choose_promoted_product_likelihood

Structural skips must be distinguished from unanswered questions. “Not applicable” for a platform attribution item is not equivalent to a missing response. The preprocessing report states that the platform field was structurally inapplicable for non-purchasers. No automatic deletion of valid extreme Likert responses was performed: a rating of 1 or 5 is a valid answer, not a data error.

The Step 2 report describes the missingness assessment as supporting MCAR. Such tests can fail to detect some patterns, so MCAR should be treated as an assumption supported by the reported checks, not as a mathematical guarantee that missingness is unrelated to all observed and unobserved values.

6. Exploratory Data Analysis

6.1 Respondent profile

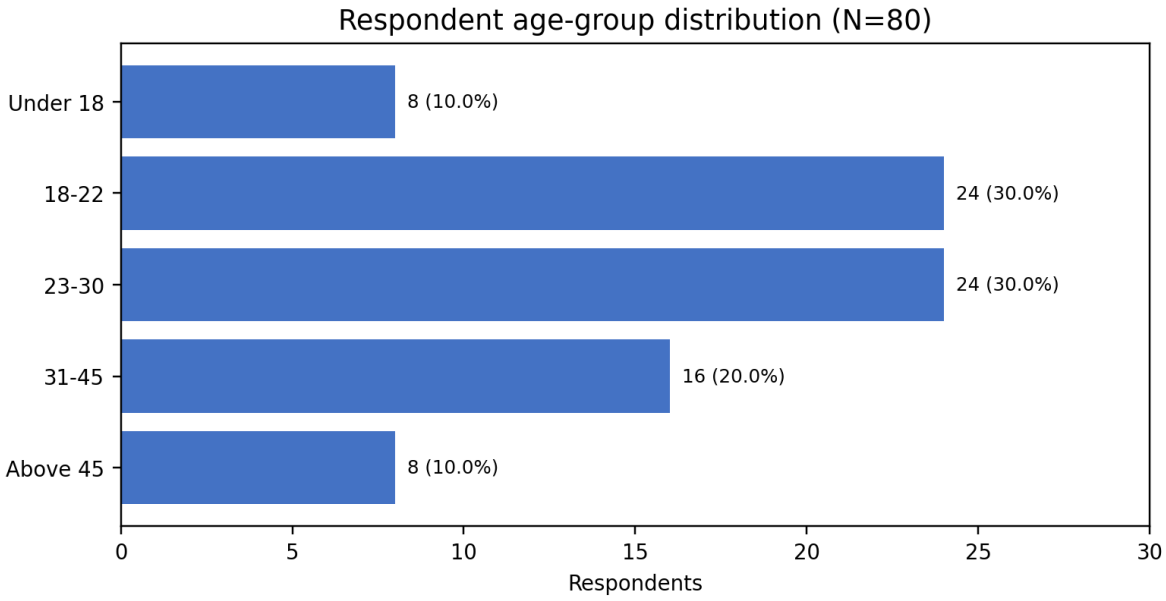


Figure 1. Respondent age-group distribution. Bars show counts and percentages of the 80 respondents.

The largest age groups are 18-22 (n=24), 23-30 (n=24), 31-45 (n=16). The sample is not necessarily representative of the wider population; age composition should be considered when interpreting the results.

6.2 Product-discovery platforms

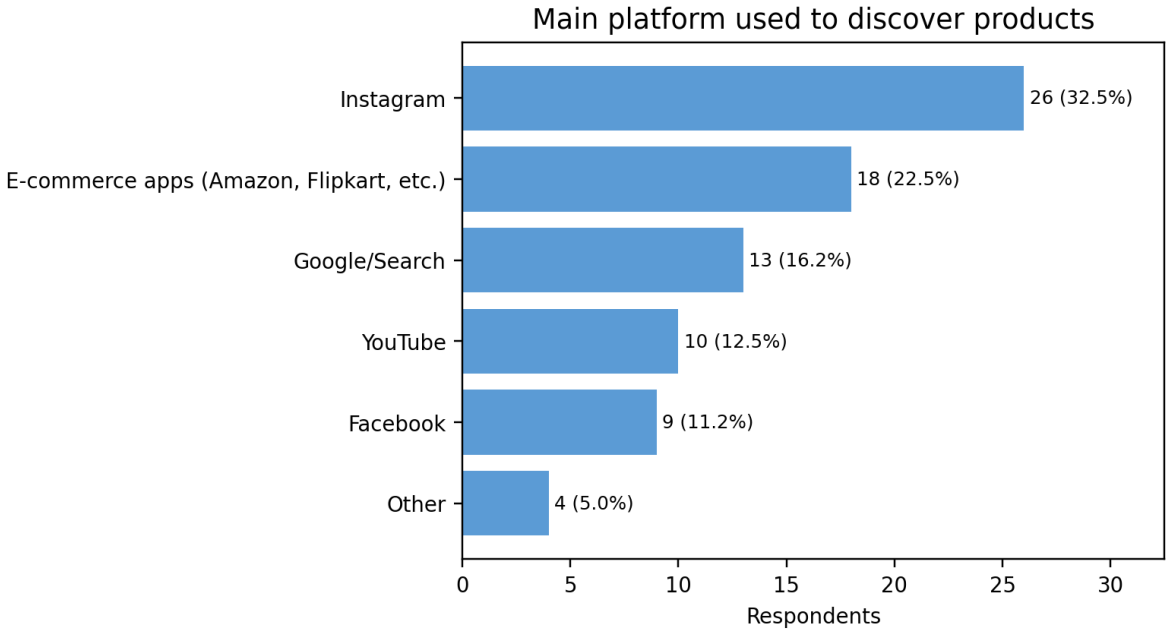


Figure 2. Main platform used to discover or learn about products.

Instagram is the most frequently selected discovery platform (26/80, 32.5%). This is a self-reported primary platform, not a measured advertising-impression share or platform conversion rate.

6.3 Preferred promotional incentives

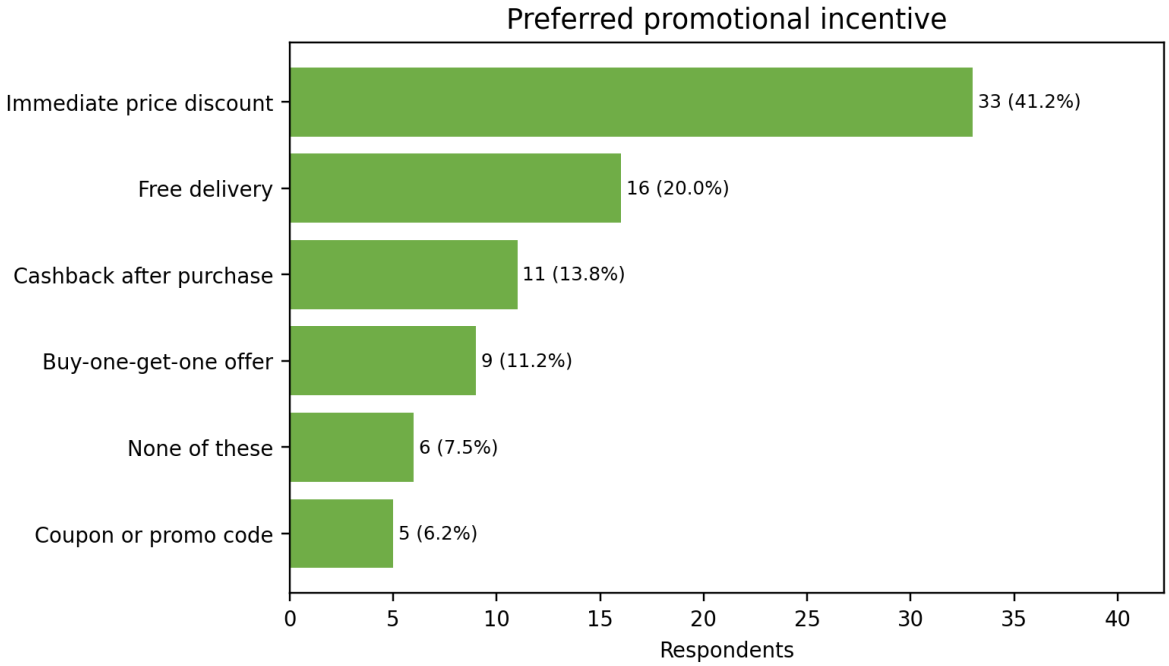


Figure 3. Preferred promotional incentive selected by respondents.

The leading preference is Immediate price discount (33 respondents; 41.2%). This is a general stated preference and should not be confused with the experimental comparison between a ₹400 discount, ₹400 cashback, and no promotion.

6.4 Reasons for ignoring advertisements

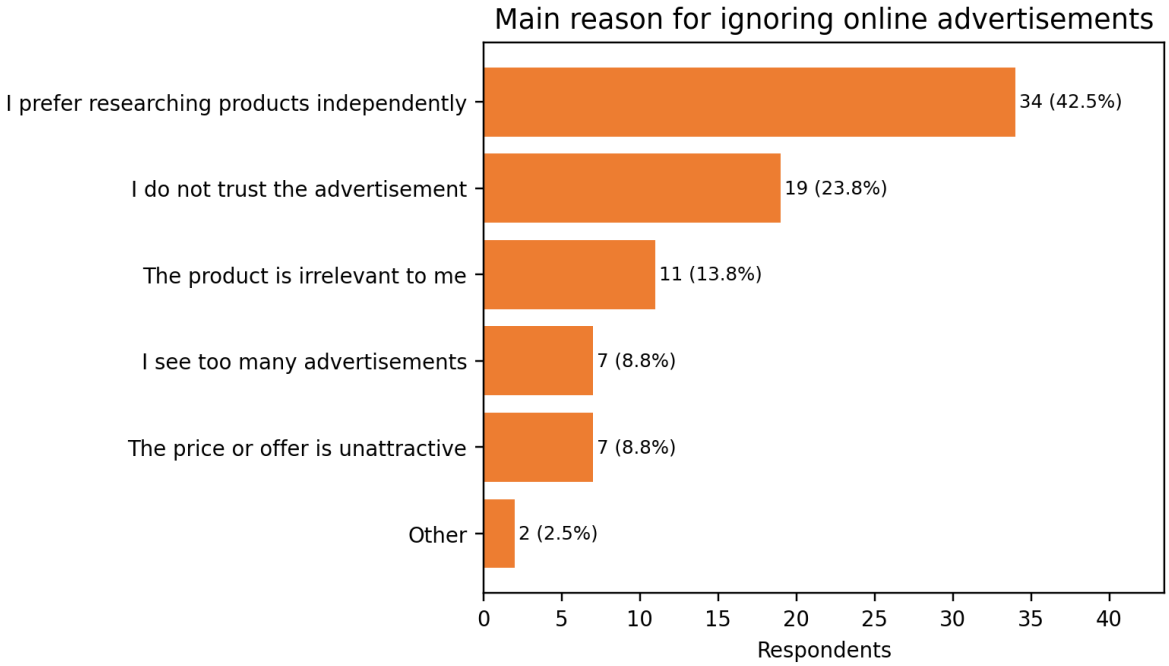


Figure 4. Main reason respondents report ignoring an online advertisement.

The most common reported reason is I prefer researching products independently (34 respondents; 42.5%). This suggests that campaign design should account for how consumers evaluate information independently, but it does not by itself establish why any specific ad was ignored.

6.5 Likert-scale rating distributions

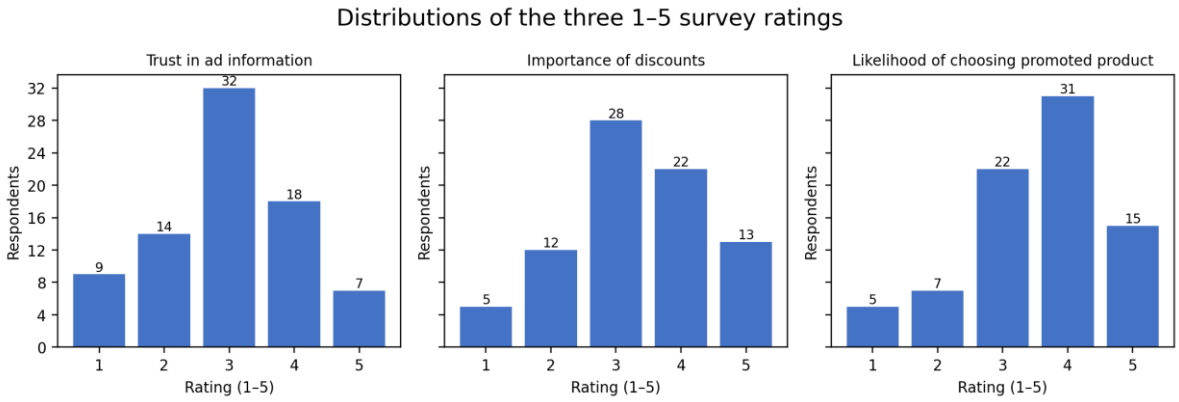


Figure 5. Full response distributions for the three 1–5 rating variables.

Rating construct	n	Mean	Median	SD	Mode
Trust in ad information	80	3.00	3.0	1.10	3
Importance of discounts	80	3.33	3.0	1.11	3
Likelihood of choosing promoted product	80	3.55	4.0	1.09	4

For trust, a rating of 3 represents the neutral midpoint; higher values indicate greater trust. For discount importance and promoted-product likelihood, higher values indicate greater importance or likelihood. For ordinal ratings, distributions and medians are useful alongside means because the numeric gaps between adjacent response options cannot be assumed to represent perfectly equal psychological distances.

7. Correlation and Outlier Analysis

7.1 Spearman correlation heatmap

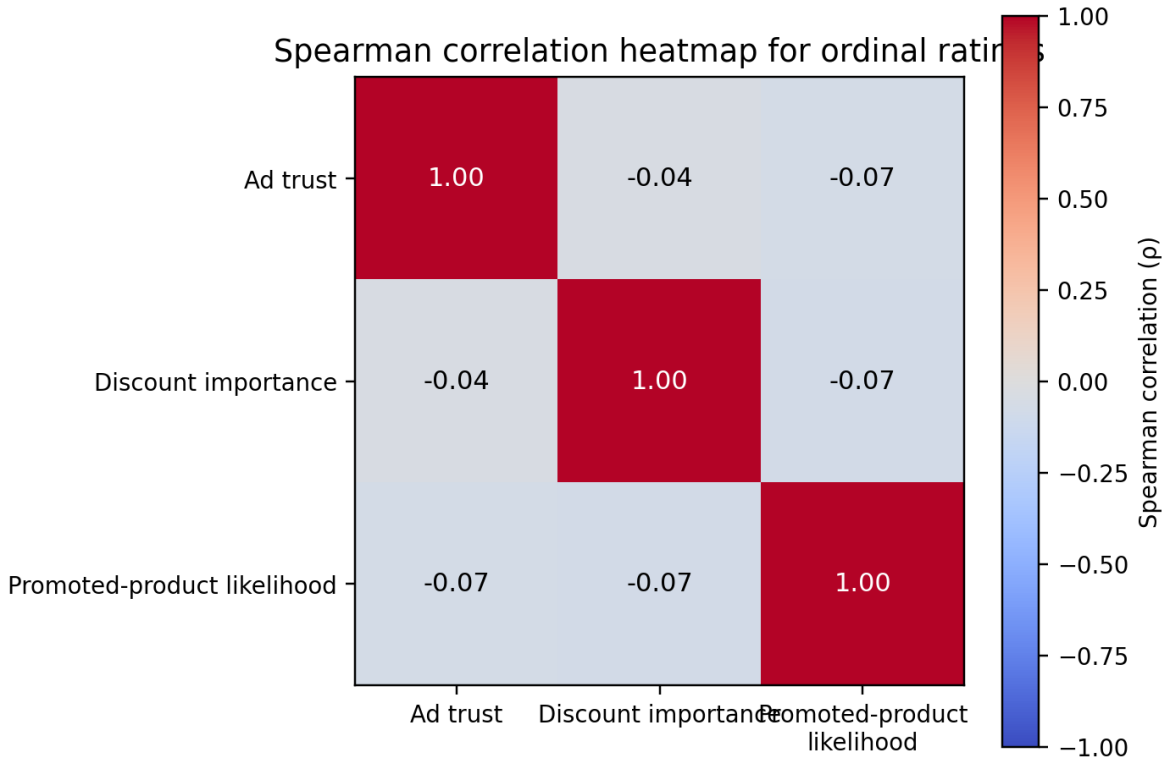


Figure 6. Spearman rank correlations between ordinal survey ratings. Off-diagonal values show the association between different constructs; the diagonal is intentionally de-emphasized.

Interpretation: Ad trust and discount importance: ρ=-0.04; Ad trust and promoted-product likelihood: ρ=-0.07; Discount importance and promoted-product likelihood: ρ=-0.07. The observed pairwise relationships are weak in magnitude. In practical terms, a respondent who gives a higher rating to one construct did not consistently give a higher or lower rating to the other in this sample. This does not establish that the constructs are unrelated in the wider population.

7.2 Box plots and IQR outlier screening

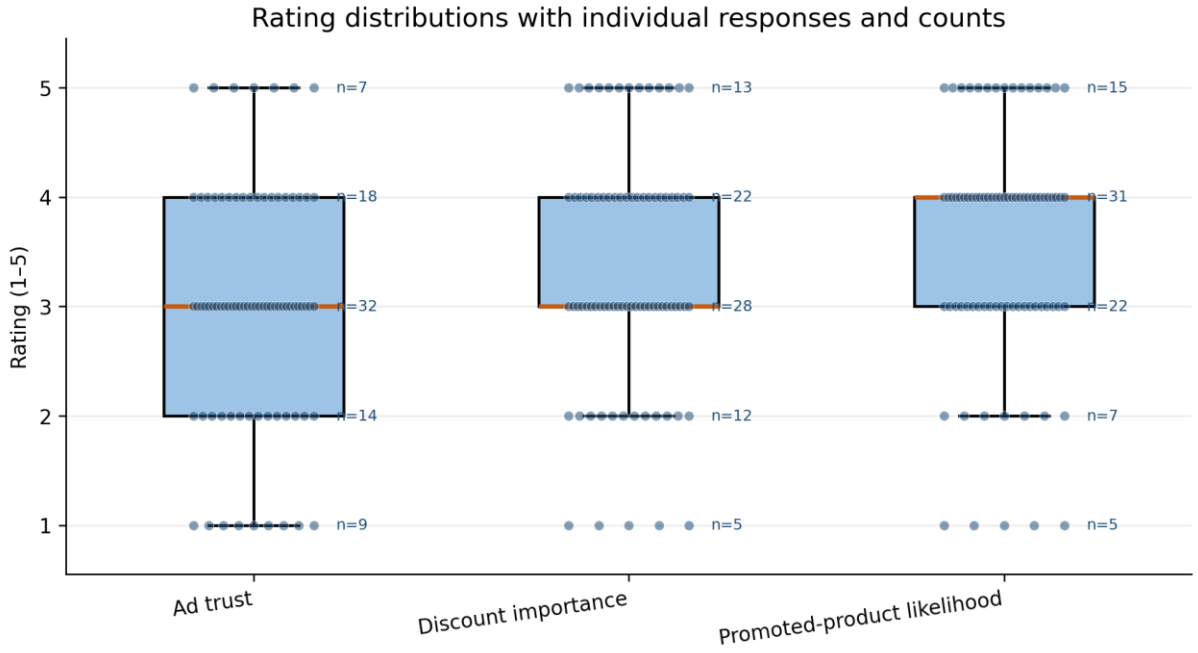


Figure 7. Box Figure 7. Rating distributions with individual responses and counts. Box plots show the median and interquartile range; individual points show the observed rating levels and their frequencies.

The IQR rule identifies values outside Q1 − 1.5×IQR and Q3 + 1.5×IQR. For bounded 1–5 ratings, the rule can flag valid ratings near the scale limits when the middle half of responses is concentrated. Accordingly, potential outliers are retained unless there is separate evidence of invalid data entry.

7.3 Main Reasons for Ignoring Advertisements

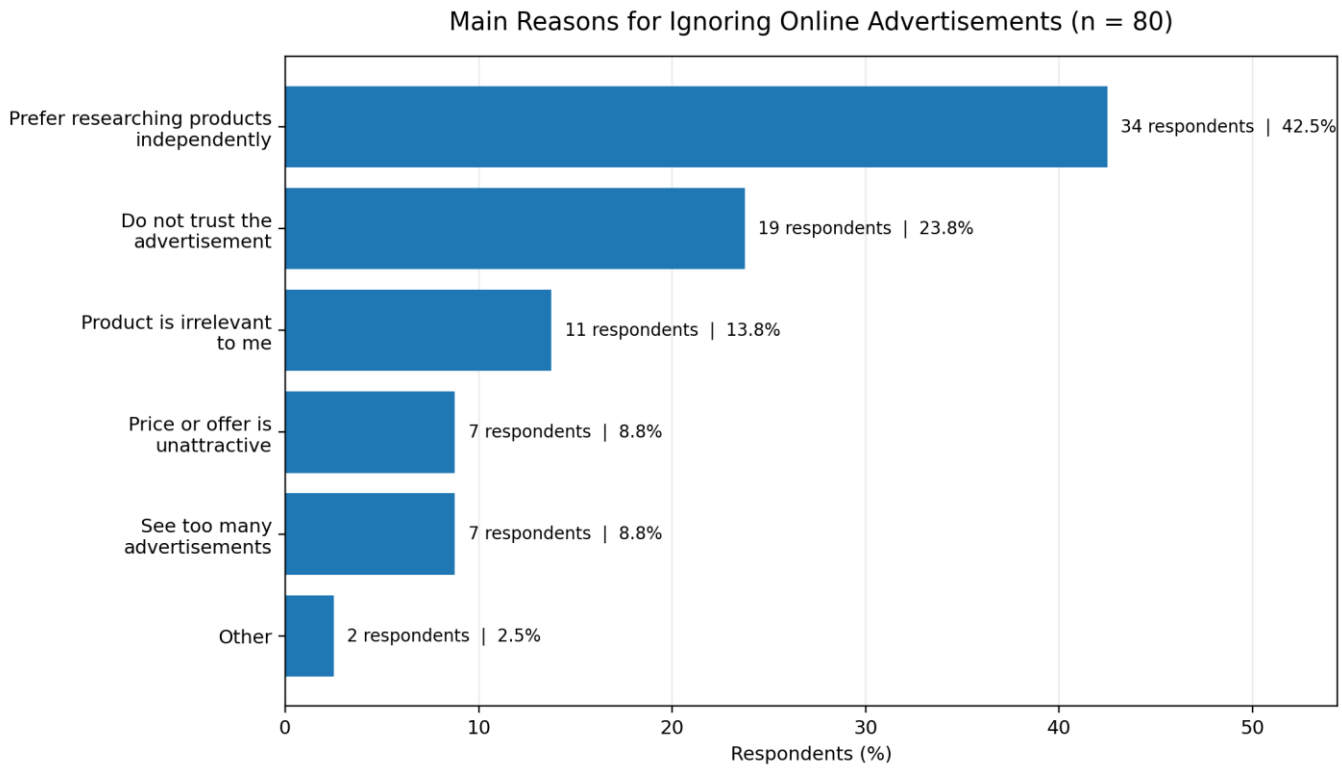


Figure 8. Horizontal bar chart showing the main reasons respondents ignore online advertisements, with the number of respondents and percentage share for each reason (n = 80).

The most common reason for ignoring advertisements is preferring to research products independently, reported by 34 respondents (42.5%). This suggests that many respondents prefer to evaluate products themselves rather than rely directly on advertising messages.

The second-largest reason is lack of trust in advertisements, selected by 19 respondents (23.8%). Together, these two reasons account for 53 respondents, or 66.3% of the sample.

Product irrelevance accounts for another 13.8%, while unattractive offers and excessive advertising each account for 8.8%.

7.4 Promotion preference across age groups

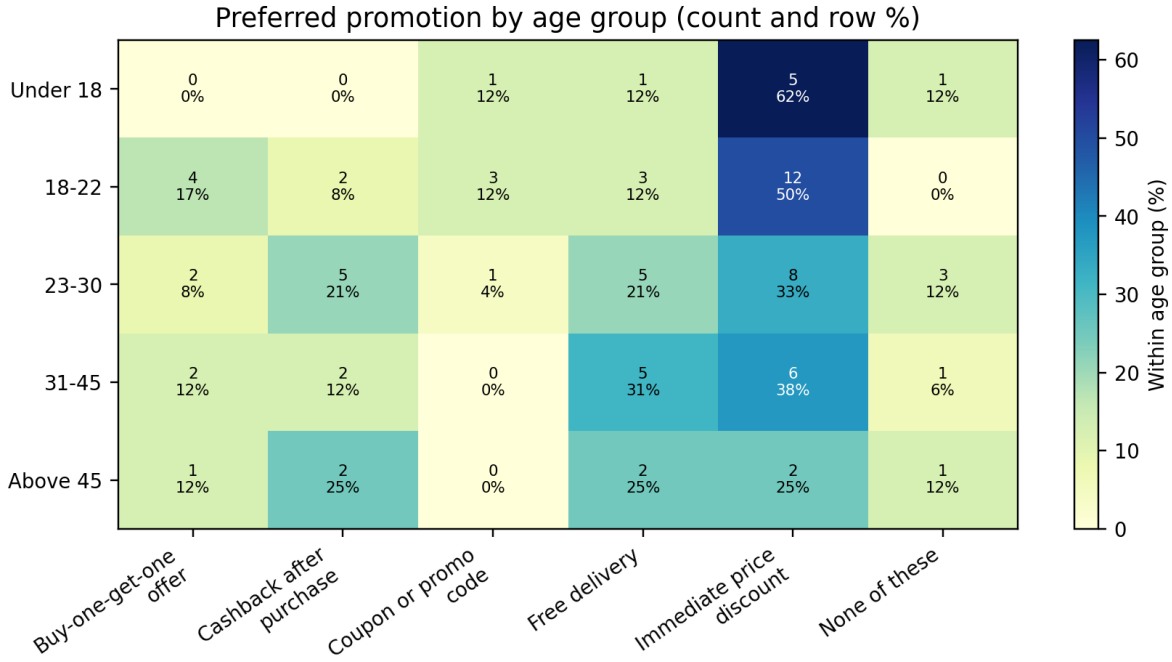


Figure 9. Preferred promotion by age group. Each cell shows the count and within-age-group percentage.

A chi-square test on the age-group by preferred-promotion table gives $\chi^2(20) = 16.09$, $p = 0.711$. Because the table has several categories and some cells may have low expected counts, this result should be interpreted cautiously; an exact or Monte Carlo method may be preferable if formal inference is required. The heatmap is primarily descriptive.

7.5 Ad frequency and reported purchase behaviour

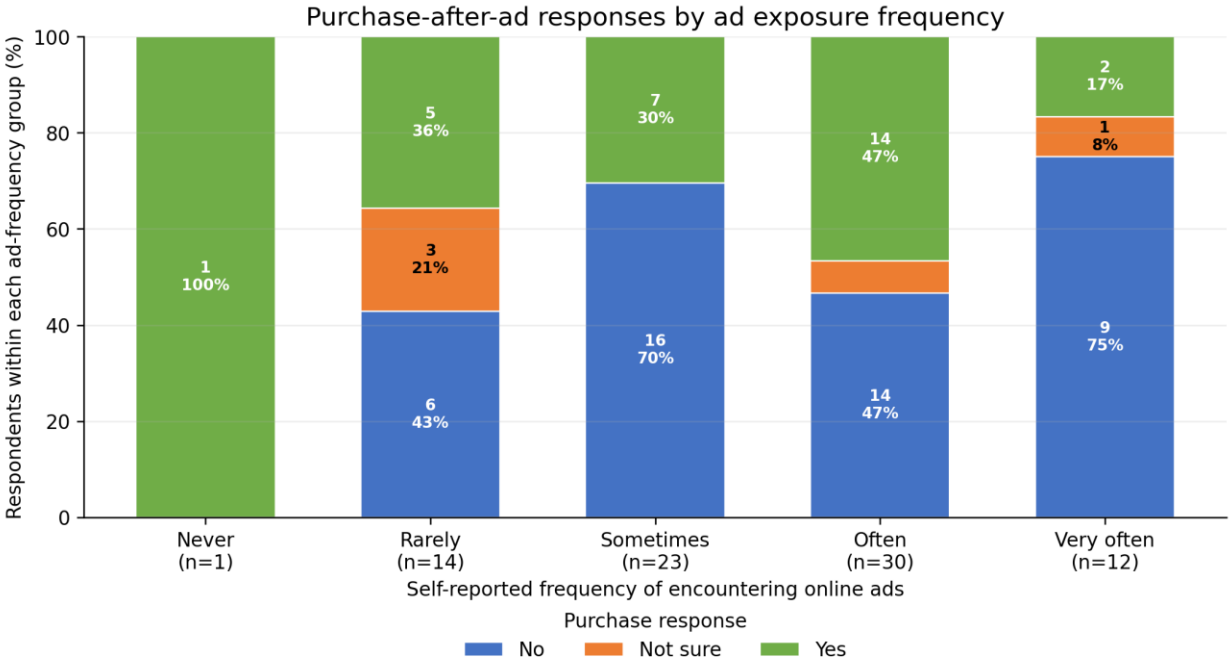


Figure 10. Purchase-after-ad responses within each self-reported ad-exposure group. Each bar totals 100%; group sample sizes and segment counts/percentages are shown to prevent small groups from being mistaken for stable rates.

This comparison describes how reported purchase responses vary across exposure-frequency categories. It does not prove that greater exposure causes purchasing, because exposure and purchase behaviour are self-reported and may be influenced by other factors.

8. Statistical Interpretation

The analysis combines descriptive summaries with association measures. The principal rating relationships are modest in magnitude in the supplied dataset; therefore, the safest interpretation is that the three attitudes are related only to the extent indicated by their observed coefficients. No causal claim is supported by these correlations.

For categorical comparisons, chi-square tests require adequate expected cell counts. Where many cells are sparse, results should be treated as exploratory and considered alongside effect sizes and category frequencies. For the promotional experiment, the appropriate primary analysis—once the scenario-level data are supplied—is a repeated-measures comparison of the 1–5 outcomes across the three conditions. A Friedman test is a reasonable ordinal starting point, followed by paired Wilcoxon tests with Holm correction if justified.

The experimental scenario outcome data were not present in the supplied cleaned CSV. Consequently, no scenario means, medians, p-values, or claim that discount/cashback/no promotion performs best is reported here. This is an intentional evidence limitation rather than a missing analysis result.

9. Recommendations

- Improve advertisement transparency: show accurate product specifications, total price, warranty and return details, and reliable review links so consumers can verify claims.
- Test promotions using the experiment responses: compare immediate discount, cashback, and no promotion using the same respondents' ratings rather than relying only on the general preference question.

- Communicate cashback clearly: show the checkout amount, cashback amount, delivery timing, and conditions. In this study, the discount reduces the upfront payment to ₹1,600, while cashback requires ₹2,000 upfront.
- Use segmentation cautiously: age and platform patterns can inform future campaign hypotheses, but 80 respondents are insufficient for strong claims about all consumer groups.
- Measure real outcomes in future work: where feasible and privacy-compliant, collect verified clicks, purchases, returns, and revenue alongside stated intention.
- Improve survey completeness: keep required questions limited to essential fields, while considering whether key outcome items should be required or offer a “Prefer not to answer” choice.

10. Limitations and Future Work

- Sample size: the dataset contains 80 respondents; subgroup percentages can be unstable and may not represent the wider population.
- Sampling bias: the recruitment method may over-represent particular ages, platforms, or consumer habits.
- Self-reported answers: respondents may misremember past behaviour, and stated intentions may differ from actual purchases.
- Optional questions: missing responses may be related to the respondent or question; missingness checks cannot prove that responses are missing completely at random.
- Experiment data are not in the supplied cleaned CSV: the relative performance of discount, cashback, and no-promotion conditions cannot yet be calculated.
- Hypothetical choices: rating an offer does not involve spending real money, so real-world behaviour may differ.
- Offer mechanics differ: cashback and immediate discounts differ in upfront payment and timing of benefit, which may influence responses.
- Association is not causation: correlation and cross-tabulation do not prove that advertising exposure or trust caused purchases.
- Generalization is limited: repeat the study with a larger, more diverse sample and, if possible, real behavioural data.

11. Conclusion

This project provides a descriptive profile of 80 respondents online advertising behaviour, advertising trust, discount importance, and promotional preferences. The most common discovery platform is Instagram, the most selected general promotion is Immediate price discount, and the leading reported reason for ignoring ads is I prefer researching products independently. The rating distributions, Spearman heatmap, box plots, and cross-tabulations provide a clear view of the sample without treating associations as causal effects. A final comparison of immediate discount, cashback, and no promotion must wait until the experiment-level response data are available.